

Experiment (9)

Spanning Tree Protocol (STP)

9.1 Objectives

To achieve importance of STP in Cisco Packet Tracker & eNSP

9.2 Equipment / Program Requirements:

Cisco Packet Tracker

eNSP

9.3 Experiment Setup:

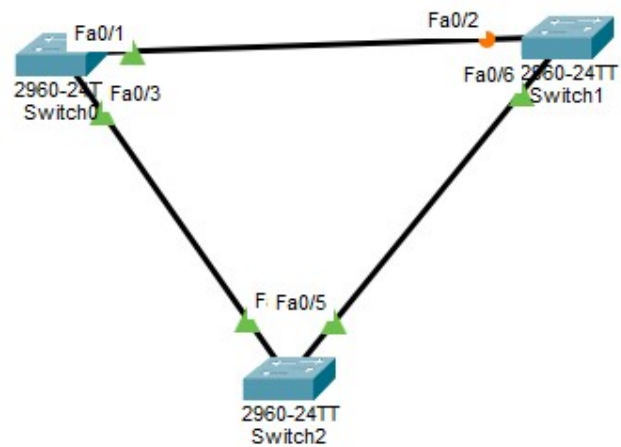
Open Simulator

Then make 3 Switches and do configurations to know Root Switch, non Root Switches, DP Ports, and BW Ports in cable.

9.4 Simulation Results:

Cisco Packet Tracker:

Topology:



SW2 is root in the First Before I force SW1 become root switch:

Switch1

Physical Config **CLI** Attributes

IOS Command Line Interface

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up

Switch>
Switch>en
Switch#sh
Switch#show s
Switch#show sp
Switch#show spanning-tree
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.64CA.68AB
Cost 19
Port 6(FastEthernet0/6)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0060.5C3D.C8A0
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Fa0/6	Root	FWD	19	128	.6	P2p
Fa0/2	Altn	BLK	19	128	.2	P2p

Switch#

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☐ Top

Sw 0:

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show sp
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
            Address     0001.64CA.68AB
            Cost        19
            Port        3 (FastEthernet0/3)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
            Address     000D.BD16.E29C
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/3          Root FWD 19        128.3   P2p
Fa0/1          Desg FWD 19        128.1   P2p

Switch#
```

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For sw1 make it root switch (forced):

Sw1(config)#spanning-tree vlan 1 priority 0

Then wait 50 second

Packet Tracer

Tools Extensions Window Help

Physical

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

Switch#

Switch con0 is now available

Press RETURN to get started.

Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#sp
Switch(config)#spanning-tree vlan pr
Switch(config)#spanning-tree vlan 1 pr
Switch(config)#spanning-tree vlan 1 priority 0
Switch(config)#

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Top

Copper Cross-Over

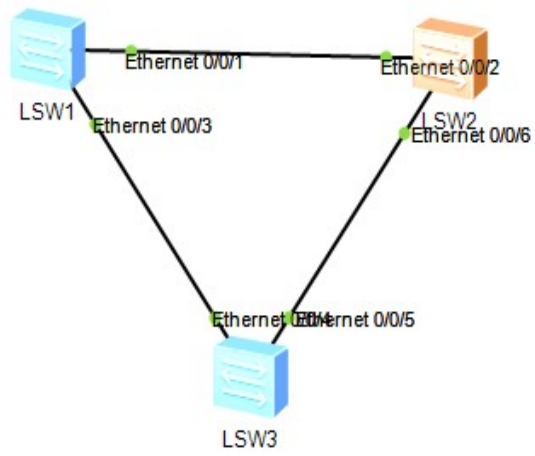
Type here to search

25°C 10:20 AM 4/25/2026

```
graph TD
    S1[2960-24TT Switch1] ---|Fa0/1 to Fa0/2| S2[2960-24TT Switch2]
    S1 ---|Fa0/6 to Fa0/5| S2
    S1 ---|Fa0/3 to Fa0/3| S3[2960-24TT Switch]
```

eNSP:

Topology:



SW1:

```
LSWI
[Huawei]stp m
[Huawei]stp mode st
[Huawei]stp mode stp
Info: This operation may take a few seconds. Please wait for a moment...done.
[Huawei]disp
[Huawei]display stp
-----[CIST Global Info](Mode STP)-----
CIST Bridge      :32768.4c1f-cc59-42c9
Config Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC   :32768.4c1f-cc43-11e0 / 200000
CIST RegRoot/IRPC :32768.4c1f-cc59-42c9 / 0
CIST RootPortId  :128.1
BPDU-Protection  :Disabled
TC or TCN received :13
TC count per hello :1
STP Converge Mode :Normal
Time since last TC :0 days 0h:0m:52s
Number of TC      :4
Last TC occurred  :Ethernet0/0/3
----[Port1(Ethernet0/0/1)](DISCARDING)----
Port Protocol     :Enabled
Port Role         :Root Port
Port Priority      :128
Port Cost(Dot1T)  :Config=auto / Active=200000
Designated Bridge/Port :32768.4c1f-cc43-11e0 / 128.2
Port Edged        :Config=default / Active=disabled
Point-to-point    :Config=auto / Active=true
Transit Limit     :147 packets/hello-time
Protection Type   :None
Port STP Mode     :STP
Port Protocol Type :Config=auto / Active=dot1s
BPDU Encapsulation :Config=stp / Active=stp
PortTimes         :Hello 2s MaxAge 20s FwDly 15s RemHop 0
TC or TCN send    :1
TC or TCN received :9
BPDU Sent         :2
TCN: 1, Config: 1, RST: 0, MST: 0
BPDU Received     :10
TCN: 0, Config: 10, RST: 0, MST: 0
----[Port2(Ethernet0/0/2)](DOWN)----
Port Protocol     :Enabled
Port Role         :Disabled Port
Port Priority      :128
```

SW2:

```
LSW2
The device is running!

<Huawei>u t l
Info: Current terminal logging is off.
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]stp mode stp
Info: This operation may take a few seconds. Please wait for a moment...done.
[Huawei]display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge      :32768.4c1f-cc43-11e0
Config Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC   :32768.4c1f-cc43-11e0 / 0
CIST ReqRoot/IRPC :32768.4c1f-cc43-11e0 / 0
CIST RootPortId  :0.0
BPDU-Protection  :Disabled
TC or TCN received :8
TC count per hello :0
STP Converge Mode :Normal
Time since last TC :0 days 0h:0m:8s
Number of TC      :8
Last TC occurred  :Ethernet0/0/6
----[Port1(Ethernet0/0/1)][DOWN]----
Port Protocol     :Enabled
Port Role         :Disabled Port
Port Priority      :128
Port Cost(Dot1T)  :Config=auto / Active=2000000000
Designated Bridge/Port :32768.4c1f-cc43-11e0 / 128.1
Port Edged        :Config=default / Active=disabled
Point-to-point    :Config=auto / Active=false
Transit Limit     :147 packets/hello-time
Protection Type   :None
Port STP Mode     :MSTP
Port Protocol Type :Config=auto / Active=dot1s
BPDU Encapsulation :Config=stp / Active=stp
PortTimes         :Hello 2s MaxAge 20s FwDly 15s RemHop 20
TC or TCN send    :0
---- More ----
```

SW3:

```
LSW3
The device is running!

<Huawei>u t l
Info: Current terminal logging is off.
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]stp mode stp
Info: This operation may take a few seconds. Please wait for a moment...done.
[Huawei]display stp
-----[CIST Global Info](Mode STP)-----
CIST Bridge      :32768.4c1f-ccfa-353a
Config Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC   :32768.4c1f-cc43-11e0 / 2000000
CIST RegRoot/IRPC :32768.4c1f-ccfa-353a / 0
CIST RootPortId  :128.5
BPDU-Protection  :Disabled
TC or TCN received :33
TC count per hello :2
STP Converge Mode :Normal
Time since last TC :0 days 0h:0m:49s
Number of TC      :6
Last TC occurred  :Ethernet0/0/5
----[Port1(Ethernet0/0/1)](DOWN)----
Port Protocol     :Enabled
Port Role         :Disabled Port
Port Priority      :128
Port Cost(Dot1T ) :Config=auto / Active=2000000000
Designated Bridge/Port :32768.4c1f-ccfa-353a / 128.1
Port Edged        :Config=default / Active=disabled
Point-to-point    :Config=auto / Active=false
Transit Limit     :147 packets/hello-time
Protection Type    :None
---- More ----
```

9.5 Conclusions:

Spanning Tree Protocol (STP) is essential in Layer 2 networks because it prevents **switching loops** that can cause broadcast storms, MAC table instability, and network failure. By creating a **loop-free logical topology**